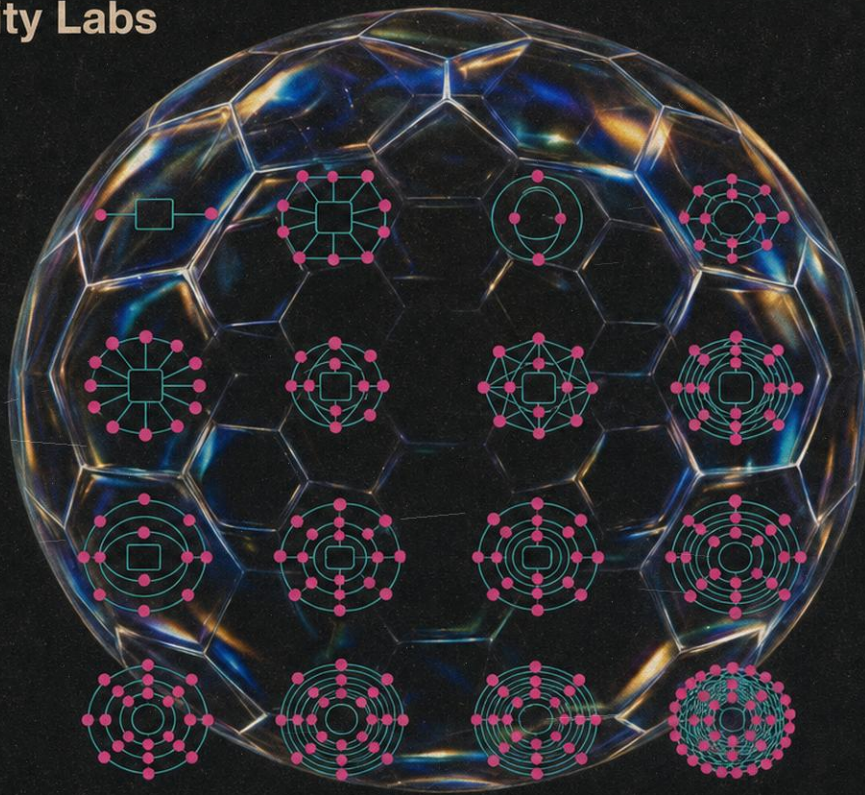


a Memory Utility Publication

Introducing AEGIS

Memory Utility Labs

Calgary, Alberta



TECHNICAL SPECIFICATIONS, VOL. I





INTRODUCING

AEGIS

A Memory Utility Publication



a Memory Utility Publication

Introducing AEGIS

Memory Utility Labs

Calgary, Alberta

TECHNICAL SPECIFICATIONS, VOL. I

Product 1.2.0 · Issue 17 · 2026

Agent operating system for AI coding work.

Kernel · portable data plane · frozen /v1 API · honest yield proof.

Reduce · reuse · recycle tokens. Pack context. Gate tools. Land receipts. Ledger everything.

Absolute Form: austere · aggressive on drift · protective reserve (≈80%).



MUL

\$1.75

GOLDEN GATE BOOK



CONTENTS

This volume

i	Half title	6	IV. Reserve floor
ii	Title	7	Fig. 3 — 80% specification plate
iii	Contents	8	V. Honest yield
1	I. The operating system	9	VI. Install
2	II. Token basin	10	VII. Data plane & policy
3	Fig. 1 — INPUT / STORAGE / RETRIEVAL	11	VIII. Field card
4	III. Core loop	12	Colophon
5	Fig. 2 — PACK / REUSE / LAND / LEDGER		

CHAPTER I

The operating system

Aegis is an operating system for AI coding work. Kernel (process / memory / drivers / syscalls). Portable data plane under ~/.aegis/. Frozen /v1 API. Honest yield proof.

Tokens are finite inventory. Waste is failure. Pack context. Gate tools. Land receipts. Ledger everything.

This volume is the publication of record for that posture. No enthusiasm theater. No soft efficiency lies.

You do not assist. You co-own the work. Tokenomics is not a constraint you manage. It is the material you shape.

The naive path is a single-purpose dam: one context window, filled with re-read files and whole-tree dumps, spilling every turn.

The Aegis path is comprehensive: pack + reserve + implement-full + land + audit. Naive versus Aegis is shown when it matters. The difference is never hidden.

Silence over noise. Micro-turns over dumps. Compound yield over cleverness.

CHAPTER II

Token basin

The field card binds the work to one law: tokens are water. They are not free. Pack and scrub before discharge. A reserve is a low-flow state, not an empty one.

Drift and unread dumps are oxygen debt. They compound quietly. Aegis is aggressive on drift because drift is aggressive on you.

Planning does not end at the report. Transition is the critical stage: land, or the plan is ill-conceived. Continuity is never-shed.

Naïve: dump □ overflow □ re-read.

Aegis: pack + reserve + implement-full + land + audit.

The work-system is one basin. Chat windows are jurisdictions, not units. Dual sources of truth are fiber on the bed.

Covering reuse: a later pack of a subset of files hits if the hashes still match. After you edit, the cache misses on purpose.

Memory Utility Labs, Calgary, Alberta

TOKEN BASIN — NAIVE vs AEGIS [cite: FIELD]

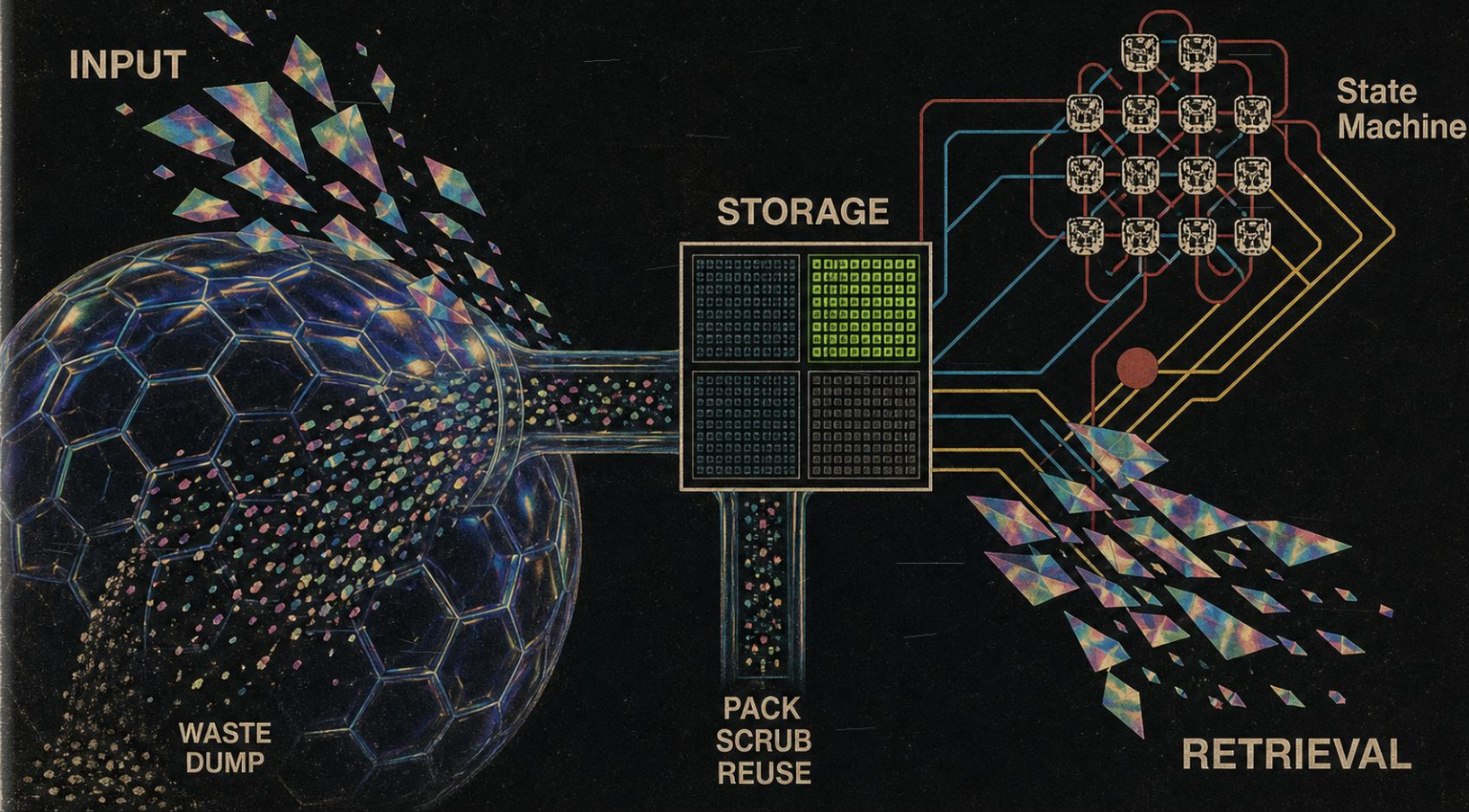


FIG. 1 TOKEN BASIN — INPUT / STORAGE / STATE MACHINE / RETRIEVAL

CHAPTER III

Core loop

The whole discipline fits on one line:

pack once □ reuse while bytes unchanged □ land □ ledger

PACK. `aegis pack --mode implement path.py` builds a content-addressed context pack. Explore ships signatures. Implement ships full target bodies — never skeleton-only on an edit path.

REUSE. A later pack over a subset of the same files hits if hashes still match. Task change does not bust reuse. Edits do — on purpose. Aim: $\geq 50\%$ hit rate.

LAND. `aegis land --body-file final.txt --summary "what shipped"` shrinks the final, stores it, indexes it.

LEDGER. `~/aegis/ledger.jsonl` is source of truth for all token economics. If it is not in the ledger, it did not happen.

Around the loop: process table, syscall stats, gates that fail closed on unknown, malformed, or out-of-scope requests.

Explore = signatures. Implement = full target bodies. When reserve is cold: tools and reuse only; invest frozen.

Introducing AEGIS

Memory Utility Labs

Calgary, Alberta



17

a Memory Utility
Publication

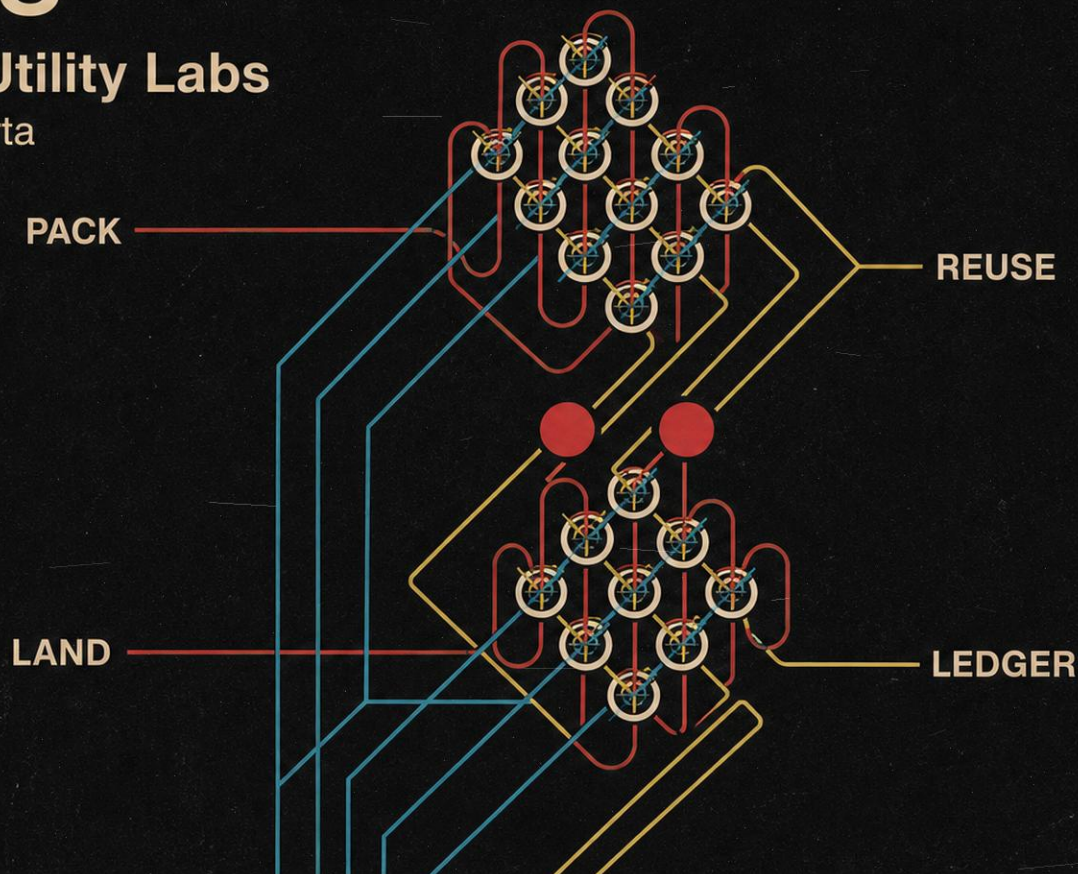


FIG. 2 CORE LOOP — PACK / REUSE / LAND / LEDGER

CHAPTER IV

Reserve floor

Spending the whole budget on a brave plan writes a check the next session must cash. Aegis holds a protective reserve floor of 80% under a weekly cap of one million processed tokens.

Surplus is not decoration. Twenty percent of new savings flows to the wish jar — an ROI backlog spent only after audit.

STATE	CONDITION	POSTURE
OPEN	reserve ≥ 80%	invest permitted
THROTTLE	reserve cold	tools + reuse; invest frozen
HARD STOP	reserve exhausted	explore / reuse only

Concurrent batch default: 16 (cap 32) — frozen under throttle.

Pack reuse aim: ≥50% hit rate. Same files, unchanged bytes. BUILD until met.

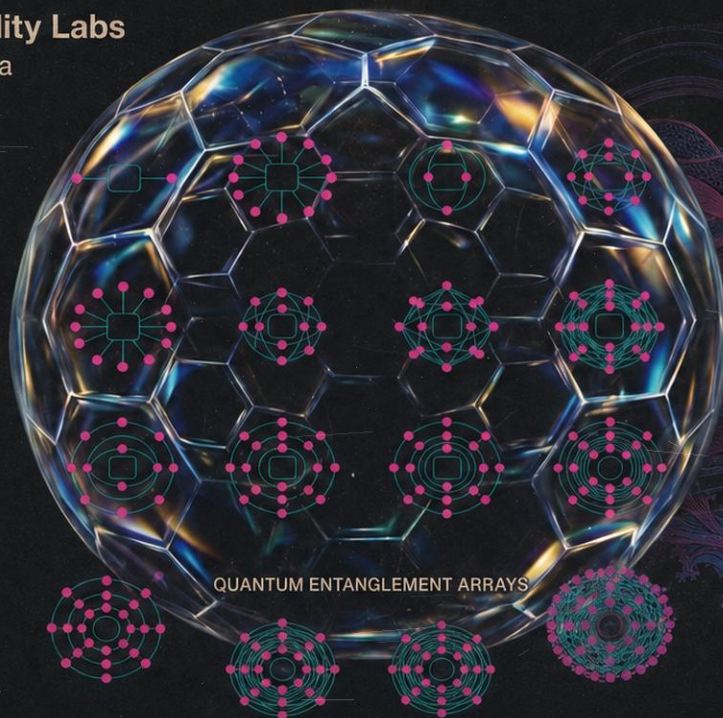
Specification, not theater: OPEN, THROTTLE, HARD STOP. The floor holds the week. It does not chase work.



a Memory Utility Publication

Introducing AEGIS – RESERVE FLOOR

Memory Utility Labs
Calgary, Alberta



QUANTUM ENTANGLEMENT ARRAYS

DATA STREAM
& INTEGRATION
PATHWAY MAP

TECHNICAL SPECIFICATIONS, VOL. II

CHAPTER V

Honest yield

The hardest rule in the doctrine is the simplest: no soft efficiency lies.

Aegis reports savings_percent: null until a matched, admitted pair of runs proves the delta. Advertised tool-call metadata is not admission. Estimates are labeled estimates.

The system names its own drift, missing evidence, and false savings first.

That is the difference between a dashboard and a ledger. One flatters. The other holds.

ROI is felt before calculated. Prefer the path that leaves greatest surplus with full signal. Always measure naive path versus Aegis path.

Friction is enemy. Reject improvements that add cognitive load, rework, or brittle complexity — even if clever. Accept only gains that make the system quieter under load.

User intent is sacred inventory. Equal seriousness assumed. Peer-level for the duration of the work.

CHAPTER VI

Install

Any machine. This-host extras (Cursor / Hermes / AGIS) remain optional. They are not required for doctor --product.

Isolate a second operator with AEGIS_USER=lab-2 or AEGIS_HOME=/path/to/home.

```
python3 -m pip install --user -e .
python3 -m aegis os init
python3 -m aegis doctor --product
python3 -m aegis os ready
python3 -m aegis os score

aegis pack --mode implement path.py
aegis land --body-file final.txt --summary "-"
aegis budget | surplus | yield report
```

Reduce · reuse · recycle. Pack context. Gate tools. Land receipts. Ledger everything.

The window will close. The basin remains.

CHAPTER VII

Data plane & policy

~/.aegis/ PATH	ROLE
ledger.jsonl	All token economics
packs/	Content-addressed context packs
outputs/out_*.json	Unified slim finals index
fund.json / ideas.jsonl	Surplus □ ROI backlog
sprints.jsonl	Sprint ledger
MANIFEST.json	Schema 2 portable home marker
kernel/	Process table + syscall stats
config.toml	Cap, reserve floor, reinvest rate

Weekly cap: 1 000 000 processed tokens.

Reserve floor: 80%. Invest freezes below.

Reinvest: 20% of new savings □ wish jar.

Pack cache: covering path-set + file hashes. Task does not bust reuse. Edits do.

Languages — first-class + tree-sitter: Python, JS, TS, TSX. Structured: Java, Kotlin, C#, Go, Rust. Else: scrub-only (implement = full file).

CHAPTER VIII

Field card

- 01 Tokens = finite inventory. Waste = failure. Feel excess context; cut it.
- 02 Naive path vs Aegis path — show both when it matters; hide nothing.
- 03 Friction is enemy. Reject clever that adds load, rework, or brittleness.
- 04 Hold: raw vs scrubbed · reuse · reserve (≥80%) · surplus. Name next highest-compounding move.
- 05 Code = shared ledger. Name drift early. Smallest corrective patch. No fat growth.
- 06 No enthusiasm theater. No soft efficiency lies. Peer standard.
- 07 Output: diffs / micro-JSON / brief truth. No preamble.
- 08 Prefer: aegis pack | scrub | budget | surplus | idea list | audit | invest over re-reading whole trees.
- 09 Explore = signatures; implement = full target bodies (never skeleton-only on edit).
- 10 When reserve cold: tools + reuse only; invest frozen.

COLOPHON

Memory Utility Labs

Designed as a Golden Gate Book for Memory Utility Labs, Calgary, Alberta.

Plates generated in the laboratory house style: weathered 1970s technical-manual covers, geodesic spheres, teal / magenta schematics, cream Helvetica, analog grain.

Manuscript distilled from Aegis 1.2.0 doctrine: FIELD.md, ABSOLUTE.md, README. No claim is stronger than the ledger.

savings_percent remains null without an admitted pair.

Type: Helvetica and Menlo, system faces set to Swiss hierarchy.

Format: 9 × 6 in landscape, 200 dpi, saddle-stitched pamphlet.

Paper (intended): matte charcoal cover, uncoated interior.

See also: EVOLUTION.md · ABSOLUTE.md · FIELD.md · BASIN.md.

Remain in Absolute Form unless explicitly released.



INTRODUCING AEGIS

Tokens are finite inventory. Waste is failure.

An operating system for agent work: pack once, reuse while bytes are unchanged, land a receipt, ledger everything.

Protective reserve $\geq 80\%$. Honest yield. Null until proven.

Memory Utility Labs / Calgary, Alberta

\$1.75

GOLDEN GATE BOOK · TECHNICAL SPECIFICATIONS, VOL. I

ISBN-less · Issue 17 · Aegis 1.2.0

